

* Teste do integral: Se f for contínuo, positivo e decrescente

Integral por partes: $u \cdot v - \int v \cdot du$

$$1) \sum_{n=1}^{\infty} \frac{1}{n+1} \rightarrow \int_1^{\infty} \frac{1}{n+1} dn = \left[\ln(n+1) \right]_1^{\infty} = +\infty \text{ diverge}$$

$$2) \sum_{n=1}^{\infty} e^{-n} \rightarrow \lim_{A \rightarrow \infty} \int_1^A e^{-x} dx = \lim_{A \rightarrow \infty} \left[-e^{-x} \right]_1^A = \lim_{A \rightarrow \infty} \left[-e^{-A} + e^{-1} \right] = e^{-1} \text{ * A série converge.}$$

$u = -x \quad du = -dx$

$$3) \sum_{n=1}^{\infty} n e^{-n} \rightarrow \lim_{A \rightarrow \infty} \int_1^A x e^{-x} dx = \lim_{A \rightarrow \infty} \left[-x e^{-x} - \int -e^{-x} dx \right] = \lim_{A \rightarrow \infty} \left[-x e^{-x} - e^{-x} \right] = \left[-\frac{\infty}{e} - \frac{1}{e} \right] - \left[-\frac{1}{e} - \frac{1}{e^2} \right] = -\frac{1}{e} \text{ * A série converge}$$

$u = x \rightarrow du = dx$
 $dv = e^{-x} dx \rightarrow v = -e^{-x}$

$$4) \sum_{n=1}^{\infty} \frac{1}{4n+1} \rightarrow \lim_{A \rightarrow \infty} \int_1^A \frac{1}{4x+1} dx = \lim_{A \rightarrow \infty} \int_{\frac{1}{4}}^{\frac{A}{4}} \frac{1}{u} \frac{du}{4} = \left[\frac{1}{4} \ln(u) \right]_{\frac{1}{4}}^{\frac{A}{4}} = \left[\frac{1}{4} \ln(A) - \ln(1) \right] = \infty \text{ * A série converge.}$$

$du = dx \cdot 4$
 $dx = \frac{du}{4}$

$$5) \frac{1}{2} + \frac{1}{5} + \frac{1}{10} \dots \rightarrow \frac{1}{n^2+1} \rightarrow \lim_{A \rightarrow \infty} \int_1^A \frac{1}{x^2+1} dx = \lim_{A \rightarrow \infty} \left[\arctg(x) \right]_1^A = \left[\arctg(\infty) - \arctg(1) \right] = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} \therefore \text{A série converge}$$

$$6) \sum_{n=1}^{\infty} \frac{1}{2n+1} \rightarrow \lim_{A \rightarrow \infty} \int_1^A \frac{1}{2x+1} dx = \lim_{A \rightarrow \infty} \int_{\frac{1}{2}}^{\frac{A}{2}} \frac{1}{u} \frac{du}{2} = \lim_{A \rightarrow \infty} \left[\frac{1}{2} \ln(u) \right]_{\frac{1}{2}}^{\frac{A}{2}} = \left[\frac{1}{2} \ln(A) - \ln(1) \right] = \infty \text{ * A série diverge.}$$

$u = 2x+1 \rightarrow du = 2dx$

$$8) \sum_{n=1}^{\infty} \frac{n}{n^2+3} \rightarrow \lim_{A \rightarrow \infty} \int_1^A \frac{x}{x^2+3} dx = \lim_{A \rightarrow \infty} \int_{\frac{1}{2}}^{\frac{A}{2}} \frac{1}{u} \frac{du}{2} = \lim_{A \rightarrow \infty} \left[\frac{1}{2} \ln(u) \right]_{\frac{1}{2}}^{\frac{A}{2}} = \left[\frac{1}{2} \ln(A) - \ln(1) \right] = \infty \text{ * Diverge.}$$

$u = x^2+3 \quad du = 2x dx$
 $dx = \frac{du}{2x}$

9)

10)

* base e série p : converge se $p > 0$ e diverge se $p \leq 1$

(11) $\sum_{n=1}^{\infty} \frac{1}{n^2}$ * série converge (12) * A série diverge (13) * série (14) * converge (15) * diverge (16) * converge (17) $\frac{1}{n^2}$ $\rightarrow$ converge

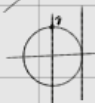
(18) $\frac{1}{n^2}$ * diverge (19) $\frac{1}{n^{\frac{100}{101}}}$ * converge (20) * converge

* base e nos próximos pedimos aplicar os testes já conhecidos, verificar o limite, caso de que verificar pelo teste do integral. Também analisar do série p .

$$21) \sum_{n=1}^{\infty} \frac{1}{n^2-1} \rightarrow \lim_{A \rightarrow \infty} \int_1^A \frac{1}{x^2-1} dx \text{ * aplicar o teste do integral} \lim_{A \rightarrow \infty} \int_{\frac{1}{2}}^{\frac{A}{2}} \frac{1}{u} \frac{du}{2} = \lim_{A \rightarrow \infty} \left[\frac{1}{2} \ln(u) \right]_{\frac{1}{2}}^{\frac{A}{2}} = \infty \text{ * A série diverge.}$$

$u = 2x-1 \quad du = 2dx$

$$22) \sum_{n=1}^{\infty} \frac{1}{n^2+1} \rightarrow \lim_{A \rightarrow \infty} \int_1^A \frac{1}{x^2+1} dx = \lim_{A \rightarrow \infty} \left[\arctg(x) \right]_1^A = \arctg(\infty) - \arctg(1) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} \text{ * A série converge}$$



$$(23) \sum_{n=1}^{\infty} \frac{1}{n^2 \sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{2.5}} \rightarrow \text{Série p que converge.}$$

$$(24) \sum_{n=3}^{\infty} \frac{1}{n^{0.75}} \rightarrow \text{Série p que diverge}$$

$$(25) \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n \rightarrow \lim_{n \rightarrow \infty} \left(\frac{2}{3}\right)^n = 0 \quad * \text{Série géométrique}$$

$$(26) \sum_{n=1}^{\infty} \frac{n}{\sqrt{n^2+1}} \rightarrow \lim_{n \rightarrow \infty} \frac{n}{\sqrt{n^2+1}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1+\frac{1}{n^2}}} = 1 \quad * \text{A série diverge}$$

* mais é geométrica de $r = \frac{2}{3}$, converge.

(27) * Série geométrica de razão $(1,075) > 1 \therefore$ diverge.

$$(28) \lim_{A \rightarrow \infty} \int_1^A \frac{1}{x^2} - \frac{1}{x} = \lim_{A \rightarrow \infty} \left[\int_1^A \frac{1}{x^2} - \int_1^A \frac{1}{x} \right] = \lim_{A \rightarrow \infty} \left[-\frac{1}{x} + \frac{\ln x}{1} \right] = \infty \quad * \text{Duas séries p separadas}$$

$$(29) \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e \therefore \text{Série diverge}$$

$$(30) \sum_{n=2}^{\infty} \ln |n| \rightarrow \lim_{n \rightarrow \infty} \ln |n| = \infty \quad * \text{Diverge}$$

(32)